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Enhanced word vector space with ensemble deep learning model for COVID-19 Hindi text sentiment analysis

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Abstract
The SARS-CoV-2 virus has spread worldwide since March 2020 and became a global pandemic. Millions of people worldwide have infected with this virus. This research work proposed an enhanced word vector space and deep learning-based hybrid model for COVID-19 Hindi text sentiment analysis. Total 36,863,393 Hindi tweets are collected for analysis. First, various natural language processing techniques are applied for text preprocessing. Next, The word vector space is enhanced by combining various word embedding techniques namely, Word2vec, Fasttext, Glove, POS2Vec, WordPosition2Vec, Lexicon2Vec, and Elmo. Finally, developed a hybrid model for sentiment classification. According to the experimental findings, highest 89.60%, 78.70%, 77.23% and 91.64% precision, recall, F-score, and accuracy, respectively, is given by proposed model. The obtained results indicate that the proposed approach is highly efficient in classifying Hindi sentiments.

Keywords Word embeddings · Glove · Fasttext · Neural networks · Text classification

1 Introduction

Every day millions of people uses social media networking website such as twitter, blogs, wikis, and others to share their opinions. The attitude and feelings of people can be analyzed by using social media messages [1]. Furthermore, sentiment analysis can also be performed using data gathered from online review platforms and customer feedback surveys. This multi-source approach allows for a comprehensive understanding of public opinion and sentiment towards a particular topic or brand. Coronavirus is highly contagious and spreads rapidly through sneezing, coughing, and speaking. Currently, it is a global issue that affects people directly or indirectly [2]. During the pandemic crisis, sentiment analysis is useful for understanding

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people's emotions, thoughts, and concerns. More effective measures can be taken to deal with the situation by analyzing of sentiments of people. Several strategies can be applied for analyzing the sentiments which are given as:

(a) Lexicon-based technique: It is a unsupervised technique in which lexical resources assign a polarity rating of each word to determine the emotion score.
(b) Machine learning model: In this model various supervised techniques are applied for sentiment classification [3].
(c) Word embedding approach: It is also commonly used technique for sentiment categorization. Various word representations in vector spaces can be learn from big text corpus using neural networks or probabilistic models for sentiment detection [4, 5]. Word embedding captures semantic and syntactic relationships between words based on text context. Word embedding is essential for many general-purpose NLP applications. It is classified into three groups: (i) Traditional word embedding namely, Word2vec and Glove, trained on enormous data and captures syntactic as well as semantic similarities [6, 7]. (ii) The second category of word embedding capture character-level sub-word properties such as Fasttext, and (iii) Elmo embedding is third category which deal with context-dependency and word polysemy [8, 9]. Due to rich corpus of English language, Elmo technique is used in various applications of NLP. There is lack of large corpus of Hindi languages [10]. Insufficient research is conducted on Hindi languages due to lack of large data-set. The single word embedding method such as Word2vec for Hindi text does not maintain the sequence of words, resulting in, the syntactic and semantic comprehension of a sentence being lost. Massive amount of text data cannot be analyzed using the single word embedding approach due to multiple meanings of same word in different sentences and phrase context. Various word embedding can be concatenated with part of speech embedding and lexicon-based approach to increase the computing capability of deep-learning method.

1.1 Objectives and motivations

Hindi sentiment analysis has unique challenges due to its rich morphology and contextual ambiguity. The combination of several word embeddings and sub-word information in word representations. The primary objective is to develop an advanced deep learning model that integrates enhanced word vector space and deep learning algorithms. This model has accurately analyzed Hindi texts from social media. Significant gaps in the existing literature on Hindi text sentiment analysis regarding the COVID-19 pandemic are filled by this manuscript.

1.1.1 This research investigates the following questions

1. Can sentiment classification accuracy in COVID-19 related Hindi text be improved by a hybrid word embedding approach compared to single word embedding methods?
2. What is the optimal combination of word embedding techniques for achieving the highest classification accuracy?
3. How does the proposed model perform compared to existing sentiment analysis methods for Hindi text?

1.2 Novelty of this work

Individual word embedding techniques like Word2vec and Glove are well-established. Our approach combines seven different embedding methods, including character-level (Fasttext)

and context-aware (Elmo) techniques, specifically for Hindi text. In the context of COVID-19, this multidimensional representation provides a richer understanding of word meaning and relationships, and potentially improved sentiment classification accuracy. There is no existing research that combines these specific embedding methods to analyze Hindi sentiment associated with COVID-19. Combining these approaches overcomes the limitations of single word embeddings mentioned in the introduction, such as the loss of syntactic and semantic information.

According to the paper's structure, Section 2 covers existing literature. Section 3 covers proposed methodology used for sentiment categorization. Section 4 consists of implementation details with result analysis. Section 5 concludes with recommendations for future research.

2 Related works

There are two basic methods for sentiment classification: dictionary-based techniques and machine learning-based techniques [11]. Vocabulary-based approaches are simple and computationally efficient [12]. Each word is associated with either positive or negative value in this approach. Lexicon-based techniques are recently integrated with machine learning methods to increase the classification accuracy. More accurate results are recorded by using machine learning techniques as compared to lexical techniques [13]. Bhat et al. analyzed approximately 80,000 tweets about COVID-19 transmission with 51.97%, 34.05%, and 13.96% positive, neutral, and negative attitude, respectively [14]. Lexical approach based 'CrystalFeel' algorithm is used by Lwin et al. to analyze more than 20,325,929 tweets regarding COVID-19 pandemic to examine public emotional responses [15]. Various feelings such as anxiety, sorrow, and happiness are analyzed by the authors. Barkur et al. analyze Indian citizens' opinions during the lockdown with word cloud and R software. They have collected 24,000 tweets by using #IndiaLockdown and #IndiafightsCorona from March 25-28, 2020 for emotion analysis [16]. Label encoding approach with SVM and logistic regression are used by Majumder et al. for sentiment identification [17]. They have used Indian tweets between March-June 2020 for analysis. Bengio et al. have used distributed representation of words with neural language model [18]. In recent years, deep learning algorithms such as Word2vec and Glove are used by various authors for word vector representation [19, 20]. The corpus size plays an important role to obtain high accuracy in this techniques. Glove technique is proposed by Pennington et al. for global context representation of words [7]. The global context is determined by statistical analysis of word occurrences. Fasttext word embedding technique with n-grams features are utilized by Joulin et al. and achieved enhanced results as compare to deep learning approaches [21]. Word embedding of 157 different languages with Fasttext approach is done by Grave et al. [22]. Authors have analyzed that accuracy rate of Hindi corpus is less as compared to 10 other languages due to its small size. Total 39 fusions of recurrent neural network (RNN), CNN, and LSTM models are evaluated by Bhargava et al. by using SAIL 2015 dataset and obtained 74.43% accuracy [23]. Hindi-English mixed coding analysis is done by Mukherjee et al. with combination of character and word data [24].

Akhtar et al. [25] include the assessment of challenges in sentiment analysis in Hindi, the creation of a benchmark setup with an annotated dataset, and the development and evaluation of machine learning models using CRF and SVM, for sentiment analysis. Model achieved 41.07% F-measure for aspect term extraction and 54.05% accuracy for sentiment classification. Their work has the following limitations.

Limtations:

- Low average F-measure.
- Moderate accuracy.
- Narrow focus on challenges of sentiment analysis in Hindi
- No further research suggestions or limitations mentioned

Pandey et al. [26] analyzed movie review sentiment in Hindi using HindiSentiWordNet for document-level polarity. According to their approach, HindiSentiWordNet is enhanced and sentiment is extracted, then the enriched lexicon is used to classify movie reviews as positive, negative, or neutral. Using an ensemble pretrained deep learning model (MVEDL), Malla et al. [27] investigate the identification of informative tweets associated with the COVID-19 pandemic. MVEDL has been achieved an impressive F1-score of 91.14% by using pre-trained transformers such as CT-BERT, BERTweet, and RoBERTa. The study also highlights real-time applications of the MVEDL model, including monitoring mental well-being, monitoring sentiment in tweets, and analyzing COVID-19 outbreak data. To analyze sentiment polarity, Imran et al. implemented a Long Short-Term Memory (LSTM) deep learning architecture with pre-trained embedding models and emotions in COVID-19 related tweets from different cultures [28]. Tweets were collected from February 2020 to April 2020. NLP-based techniques were used to train the sentiment and emotion analysis model. Neighboring countries within a region displayed strong sentiment correlation in tweets. USA, Canada, Pakistan, and India exhibited similar polarity trends, while Norway and Sweden exhibited different ones. A multimodal approach has been proposed by Borah et al. to analyze Indian tweets regarding the hesitancy toward the COVID-19 vaccine, based on both content and network features [29]. This approach uses transformer BERT and graphBERT models to integrate multiple data representations and demonstrate improved classification results over current state-of-the-art approaches. The study shows that vaccine hesitancy in India initially peaked after the program launch, then gradually decreased until August. In December, it reached its lowest level. A slight increase in hesitancy was observed in August, which gradually increased until November. The lowest hesitancy was observed in December. The opinions and sentiments of Indian Twitter users towards COVID-19 vaccination were evaluated by vishwakarma et al. [30]. Using relevant hashtags and keywords, tweets were collected between January 2021 and March 2023. Using Natural Language Processing techniques, sentiment analysis of Twitter data was conducted to determine perceptions and outcomes of the COVID-19 vaccination. According to the analysis, positive and negative sentiment varies across demographic groups. As compared to other ML models, such as SVM and Random Forest, LSTM had the highest accuracy (88.5%). Mittal et al. [31] presented an approach to enhance the coverage of the Hindi SentiWordNet and investigating the impact of negation and discourse rules for Hindi sentiment analysis. The proposed method achieved an accuracy of 80.21% in classifying positive and negative reviews in Hindi language content. The study addresses the need for analyzing Hindi language content to gain insights into the opinions expressed by people and various communities. Gobbo et al. proposed GradeAid, a novel ASAG framework employing state-of-the-art regression to analyze lexical (TF-IDF) and semantic (BERT Cross-Encoder) [32] features of short answers, and converting them into a similarity score [33]. It achieved root-mean-squared errors as low as 0.25 on specific datasets and questions. Topbas et al. [34] utilized the Twitter web API to obtain a dataset of COVID-19-related tweets with manually assigned sentiment labels, and applied deep learning models (RNN and BERT) for sentiment analysis. The experimental results demonstrated the effectiveness of both models. The RNN and BERT models achieved 86.4% and 83.14% accuracy respectively. Salma et al. [35] perform sentiment analysis to understand public opinion about Peduli Lindungi. Various

classification methods were used in their research, which include K-Nearest Neighbors, Support Vector Machines, and Naive Bayes Classifiers. These models achieved the accuracy of sentiment classification based on SVM was 76.5%, NBC was 72.3%, and KNN was 59.1%. Based on CNN, LSTM, and multi-input CNN, Tummalapalli et al. evaluated the accuracy of classification of Hindi and Telugu texts. [36]. They have captured the morphological distinctions of Hindi language at both word and character level. Summary of some recent related work is shown in Table 1.

Limitations and challenges of existing techniques Sentiment analysis is an area of NLP that automatically identifies the sentiments conveys with words. Significant works has done by various authors in this field. Limitations and challenges of existing works are given as:

- Subjectivity and contextual dependence: Sentiment analysis depends on the speaker's tone, cultural background, and specific context. In this case, existing techniques are not able to capture nuances of emotion accurately.
- Language and domain dependence: Sentiment analysis model trained on a specific language or domain may not be generalize for other languages or domains. For example, a model trained for movie reviews may not perform well on product reviews.
- Sarcasm and irony: It is sometimes difficult to detect sarcasm and irony of text. The sentiment expressed by the Sarcasm text may be different from literal meaning. In such cases, existing techniques are unable to identify sentiments accurately.
- Ambiguity and polysemy: Sentiment analysis can be challenging with words having multiple meanings. Meaning of these words are context dependent. It can lead to ambiguity in text sentiment analysis.

3 Proposed model

The proposed methodology for COVID-19 Hindi text sentiment analysis includes various steps such as (i) Data gathering and preprocessing, (ii) Composition of word embedding techniques, and (iii) Sentiment classification based on hybrid deep learning model. Block design of proposed work is illustrated in Fig. 1. Each phase of the proposed work is discussed briefly in the following subsections.

3.1 Data collection

First, Hindi tweets with #coronaindia, #lockdownindia, #staysafe, #stayhome, #indiafightscorona, and #covid19 #coronavirus, hashtags are gathered using Twitter API. All hashtags used for tweet collection are shown in Table 2. Total 36,863,393 COVID-19 Hindi tweets are collected through Tweepy API. Collected dataset contains highly dense text and many textual characteristics.

Table 1 An overview of recent related research

Reference	Approach/Model	Accuracy
Grave et al. [22]	Fasttext	66.7%
Bhargava et al. [23]	CNN and LSTM	74.43%
Garg et al. [37]	Lexicon-based	84.2%
Hussaini et al. [38]	Score-based	87.4%

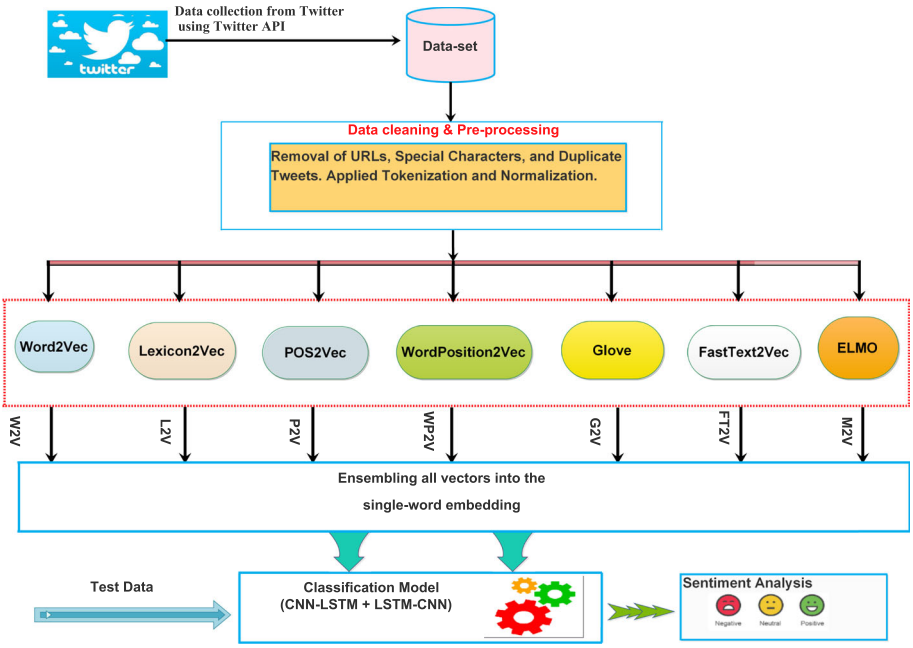


Fig. 1 Proposed model based on enhanced word vector space

3.2 Data preprocessing

It is performed as collected tweets are unorganized, unlabeled, informal, and chaotic. This process includes data cleansing, tokenization, stop word deletion, and data labelling steps as shown in Fig. 2. It decreases required memory space and accelerates further data processing.

1. **Data cleaning:** Regular expressions are used for data cleaning. This process includes removal of extra words, HTML elements, emojis, and data numbering patterns. It also removes extra characters from sentences, emails, URLs, and punctuation.
2. **Tokenization:** Tokenization is performed by using Indic NLP library to partition the long texts into small chunks known as tokens [39, 40].
3. **Removing stop word** Any useful information can not be extracted with stop words. So, all stop words are eliminated. Stop word dictionary contains 576 words for Hindi text as shown in Fig. 3 [41].

Table 2 Keywords used to crawl the tweets

Twitter hashtags	Total number of tweets
#covid19	4,637,295
#coronaindia	7,592,476
#lockdownindia	3,864,891
#staysafe	6,985,117
#stayhome	5,874,106
#indiafightscorona	3,541,705
#coronavirus	4,367,803

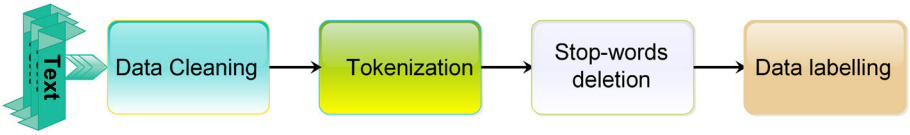


Fig. 2 Pre-processing steps

4. **Data labelling** The HindiSentiWordNet (HSWN) is used to label and extract sentiments from phrases of input dataset [42]. This lexicon comprises Hindi emotional terms based on the polarity.

3.3 Word embedding

It is a technique used to represent words as vectors of real numbers in a high-dimensional space, and each dimension corresponds to a particular feature of word. Word embedding aims to capture the semantic and syntactic relations between words in a given context. There are several types of word embeddings techniques are exists. Some of the most common techniques are discussed in the following subsection. Proposed word embedding technique



Fig. 3 List of Hindi stop word

contains seven techniques namely, Word2vec, Fasttext, Glove, POS2Vec, WordPosition2Vec, Lexicon2Vec, and Elmo. Each technique is briefly described in the subsequent subsections.

3.3.1 Word2Vec

It is a neural network approach developed by Mikolov et al. [4]. A large dataset is required to train this technique and it gives high-dimensional vector representations of each word. It employs Continuous Bag of Words (CBOW) model and skip-gram architectures to predict target word. Prediction is independent of words sequence in CBOW architecture. Mathematical description of this technique is given as:

$$T_\theta = \frac{1}{X}\sum_{y=1}^{X} \log T\left(a_y \mid a_{y-m}\cdots a_{y-1}a_{y+1}\cdots ..a_{y+m}\right) \tag{1}$$

here, T_θ denotes probability of target word, m indicates how many words precede and follow the target word. a_y denotes target word at each timestamp y. This model accept context window of size m at every timestamp y with target word a_y.

3.3.2 FastText

It is fast, free open-source package for text representation and classification [21]. In contrast to embeddings, this technique can provide embeddings for misspelled words, unusual words, and untrained words [4]. This model is pre-trained with Facebook data. Total 309 terms of tweeter Hindi dataset applied for present work is missing. So, Genism FastText algorithm is used for training on Hindi dataset to determine word vectors.

3.3.3 Glove

It is utilized to construct low-dimensional distributed word vectors [43]. It reduces dimensionality of the word-context matrices by taking record of phrase paired co-occurrences in the dataset. It presents an attenuation function to compute weight based on gap among two or more words in context window [44].

3.3.4 ELMo

It is context based character and word-level technique which considers complete sentence while determining the appropriate embedding for each word [45]. It uses bi-directional RNN to generate embeddings which depends on next and previous words of input sentence [46]. This technique contextualizes each token by concatenating BiLSTM states.

3.3.5 POS2Vec

It is used to describe the syntactic structure of words. Word POS information is vital in sentiment analysis for emotion identification. Words with distinct POS tend to have diverse meanings and feelings. A POS-tag embedding is comparable to word embedding. Each POS tag has a constant vector integrated with word. This enables the model to recognize the structure of sentences. A TnT-based Hindi POS tagger is deployed to tag the Hindi dataset [47].

3.3.6 Lexicon2Vec

It is polarity-rated phrases and words based lexicon. It can be applied to any text pieces for analysis. Every lexicon contains words with its associated emotion weights. Several sentiment lexicons are available and selection of right lexicon is critical. Some lexicon combinations are more accurate than others. Following lexicons with allocated vector for each word is used in this work.

- Hindi SentiWordNet (HSWN) [48].
- Utilizing WordNet Graph Traversal for Hindi Subjective Lexicon Generation [49].

3.3.7 WordPosition2Vec

It is a method of generating word embeddings, which are numerical representations of words that capture their semantic meaning. In this method, every phrase is mapped as a vector in an area of high dimension. By considering these factors, the resulting embeddings are better able to capture the nuances of language.

3.3.8 Proposed enhanced word vector space(Ewvs)

Multiple-word embedding techniques can be an effective way to enhance the accuracy of classification model. In the proposed work, word2Vec, ELMo, GloVe, and FastText embeddings are combined to improve classification accuracy. After data pre-processing, load all four Word2Vec, Elmo, Glove, and Fasttext techniques with dimensions 200, 1024, 100, and 300, respectively. Next, find the vector of each word given in the vocabulary by applying following steps:

(i) Retrieve word vector generated by word2Vec embedding. If it is unavailable, assign 0 and store it in a vector.
(ii) Retrieve word vector generated by Elmo embedding. If it is unavailable, assign 0 and store it in a vector.
(iii) Retrieve word vector generated by Glove embedding. If it is unavailable, assign 0 and store it in a vector.
(iv) Retrieve word vector generated by Fasttext embedding. If it is unavailable, assign 0 and store it in a vector.
(v) Finally, concatenate all vectors as a single vector of size 1624.

This concatenated embedding captures more comprehensive representation of the given text as compared to individual embedding. Further, concatenated embedding is given as input to classification model for training.

3.4 Classification model

Hybridization of CNN and LSTM is developed for Categorization of sentiment. The proposed hybrid model is illustrated in Fig. 4.

1. **CNN-LSTM model:** The hybrid model of CNN-LSTM receives embedded words as an input via convolutional layer. The output CNN model with reduced dimension is given as input to LSTM layer for further computation. The CNN model captures the local context information while LSTM conducts model learning.

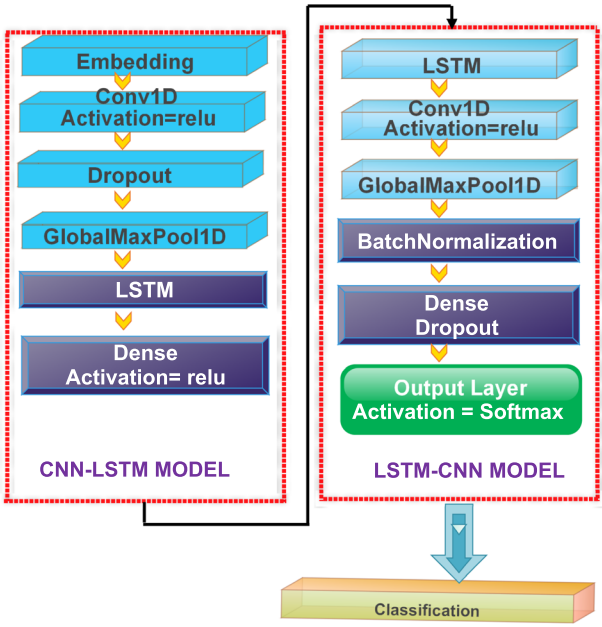


Fig. 4 Architecture of hybrid proposed model

2. **LSTM-CNN model:** The LSTM and CNN models have their own distinct features and both can be combine to get better outcome. LSTM is able to learn long-term dependencies while CNN extract characteristics from the data set. First, LSTM layer embeds each token and generates different representation of given input. Outcome of LSTM layer is utilized by CNN layer to extract local features. Output of convolution layer is combined as single dimension and assigned sentiments either as positive, negative, or neutral.
3. **CNN-LSTM + LSTM-CNN model:** The hybridization of CNN-LSTM and LSTM-CNN model is shown in Fig. 4. It is composed of the following layers:

   - **Input layer:** This layer specifies the input shape of the hybrid model. It accepts word embeddings as input. Input shape is given as (max len, embedding dim), where max len denotes the maximum length of words, and embedding dim denotes the dimension of embeddings words.
   - **Convolutional layer:** It extracts local features from the input sequence. A total of 64 filters with kernel sizes of 5 and Relu activation function are defined in this paper.
   - **Pooling layer:** This layer minimized the convolutional layer output. A pool size is given as 5 in this work.
   - **LSTM layer:** This layer captures temporal dependencies between input sequences. This study used 64 LSTM units with a dropout rate of 0.2.
   - **Dense layer:** This layer maps the previous layers' output to the desired output size of 1.The sigmoid set as an activation function in this layer.

4 Experimental setup

Total 36,863,393 Hindi tweets are collected from 2020 and 2021 to analyze people sentiment towards COVID-19.

4.1 Data preprocessing results

Tokenization process is performed after data cleaning in data preprocessing step. An example of extracted tokens from the original text data is given below:
Sample of original tweet

ब्रिटेन से दिल्ली लौटे 38 लोग Covid-19 संक्रमित सात में वायरस का नया स्ट्रेन मिला

Extracted tokens from sample tweet

ब्रिटेन, से, दिल्ली, लौटे, 38, लोग, Covid-19, संक्रमित, सात, में, वायरस, का, नया, स्ट्रेन, मिला

4.2 Results of sentiment analysis

The classification result of tweets based on emotional assessment is depicted in Table 3. Total 17,558,034; 11,564,047; and 7,741,312 tweets are classified as neutral, positive, and negative respectively. Table 4 shows the month-by-month distribution of collected datasets. The monthly sentiment analysis is shown in Fig. 5(a), (b), (c), (d), (e), and (f). For month of Mar-Apr, 20% (1028,137), 39% (2,004,867), and 41% (2,107,681) sentiments are analyzed as positive, negative, and neutral, respectively. Total 30% (1,294,376), 20% (862,918) and 50% (2,157,293) are noted as positive, negative, and neutral, respectively, in months of May-Jun. During Jul-Aug, 12% (797,638), 32% (930,578), 56% (3722,313) responses are analyzed as positive, negative feelings, and neutral, respectively. In Sep-Oct, 27% (2,094,659), 35% (2,715,299), and 62% (4,809,959) are analyzed as positive, negative, and neutral sentiment, respectively. From Nov-Dec, 31% (1,437,710), 21% (973,932), and 48% (2,226,132) responses are positive, negative, or neutral, respectively. Between Jan-Feb, 30% (2,509,607), 32% (2,676,914), and 38% (3,178,836) are analyzed as positive, negative, and neutral sentiments, respectively. WordCloud modules which visualised words related to each sentiment are illustrated in Fig. 6.

4.3 Classification results

Sentiment classification results of ensemble hybrid model along with effectiveness of each classifier is evaluated by computing F1-score, recall, precision, and accuracy. These parameters are defined as follows:

$$Accuracy = \frac{\sum(V_H, V_A)}{\sum(V_H, P_H, V_A, P_A)} \tag{2}$$

Specifically, V_H, V_A, P_H, and P_A represent accurate identification, misidentification, accurate rejection, and erroneous rejection, respectively.

$$Precision = \frac{(V_H)}{\sum(V_H, P_H)} \tag{3}$$

Table 3 Sentiment distribution of given dataset

Total tweets	Neutral	Positive	Negative
36,863,393	17,558,034	11,564,047	7,741,312

Table 4 Month wise number of tweets

Month	Number of tweets
Mar-Apr	5,140,685
May-Jun	4,314,587
Jul-Aug	6,646,988
Sep-Oct	7,757,999
Nov-Dec	4,637,776
Jan-Feb	8,365,358

$$Recall = \frac{(V_H)}{\sum(V_A, P_A)} \tag{4}$$

$$F1 - score = \frac{2 * (Recall * Precision)}{\sum(Recall, Precision)} \tag{5}$$

4.4 Parameters tuning

Table 5 shows the values given to different parameters of hybrid models.

- **Batch Size:** Different batch sizes (32, 128, 256) were employed to evaluate their effects on model convergence and efficiency.
- **Epochs:** Varied epochs (175, 143) to ascertain model stability and convergence over training cycles.
- **Optimizer:** Utilized Adam optimizer consistently across all model configurations.

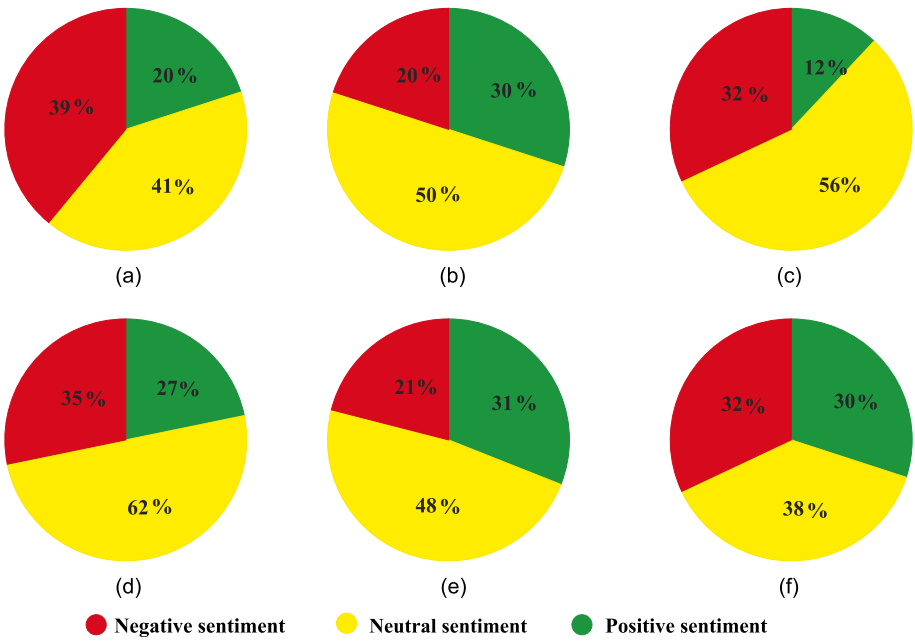


Fig. 5 Month wise sentiment distribution



Fig. 6 Most frequently words used as negative, positive, and neutral

- **Activation Functions:** Experimented with SGD, Rrlu, and Relu activations for evaluating model non-linearity and performance.
- **Dense Layer Sizes:** Tested diverse sizes (44, 64, 120) for the dense layer to determine optimal network complexity.
- **Pooling Layer Configurations:** Investigated 'same' padding and a max-pooling layer size of 2 for pooling layer consistency.
- **Learning Rates:** Ensured uniform learning rates of 0.001 for fair model comparison.
- **Dropout Rates:** Explored dropout rates (0.2, 0.5) for regularization effects and model generalization.
- **Filters and Kernel Sizes:** Varied filter counts (32, 64) and kernel sizes (5, 8, 12) in convolutional layers to evaluate feature extraction capabilities.

All models are properly calibrated with optimum parameter values. The embedding layer's vector dimension is set as 400. The hidden layer is given as 120 neurons. The learning rate with Adam optimizer is configured as 0.001. A 32-neuron layer is used to prevent overfitting with 0.5 dropout layer. The output layer is optimised with softmax activation function. The unique hyper-parameter values are given to both CNN and LSTM models.

Table 5 Parameter values given to all three Model

Parameter name	CNN- LSTM	LSTM- CNN	CNN-LSTM+ LSTM-CNN
	Value	Value	Value
Batch size	32	128	256
Epoch	175	143	143
Optimizer	Adam	Adam	Adam
Activation function	SGD	Rrlu	Relu
Dense layer	44	64	120
Pooling layer padding	Same	Same	Same
Max-pooling layer size	2	2	2
Momentum	0.7	0.7	0.7
Learning rate	0.001	0.001	0.001
Dropout	0.2	0.5	0.5
Filters	32	64	64
Kernel size	5	8	12

Table 6 Accuracy comparison between proposed model and existing work

Reference	Approach/Model	Accuracy
Bhargava et al. [23]	CNN and LSTM	74.43%
Hussaini et al. [38]	Score-based	87.4%
Garg et al. [37]	Lexicon-based	84.2%
Grave et al. [22]	Fasttext	66.7%
Madani et al. [50]	Random forest	79%
Didi et al. [51]	FastText+with TF-IDF	88.72%
Proposed model	Enhanced word vector space	91.64%

4.4.1 K-fold cross-validation

The proposed model has been evaluated using stratified k-fold cross-validation. The data is divided into k equal folds, where k-1 fold is used for training and the rest for testing. It is repeated k times, ensuring that each data point is used for both training and testing. The selection of k = 10 ensured a robust evaluation across the entire dataset. There are several reasons why we chose cross-validation over a simple train-test split:

1. Reduced variance: Cross-validation reduces accuracy estimates variance compared with a single train-test split.
2. Improved robustness: Using multiple folds allows us to evaluate the model more often and decrease the risk of overfitting.
3. Data utilization: Cross-validation uses all the data for both training and testing to maximize information.

4.5 Comparison with existing model

A comparison of the suggested model and current existing work is presented in Table 6. Bhargava et al. obtained 74.43% accuracy for language identification with CNN and LSTM models. Hussaini et al. achieved 87.4% accuracy by using score-based language identification. Highest 84.2% accuracy is obtained by Garg et al. with lexicon-based classifier. Grave et al. obtained 66.7% accuracy for multi-language identification with FastText word embedding. During the Covid-19 pandemic, Madani et al. provided a method for detecting fake news on Twitter. This study combines NLP, machine , and deep learning techniques and obtained 79% accuracy. An early warning system based on supervised machine learning has been developed by Didi et al. Different features are extracted by using TF-IDF, and Word2Vec in

Table 7 Evaluation comparison between proposed and deep learning model using Word2vec embedding

Word embedding	Models	Precision	F-score	Recall	Accuracy
	CNN	77.45%	70.25%	71.63%	71.46%
	LSTM	78.65%	69.78%	70.47%	78.72%
Word2vec	CNN+LSTM	80.23%	79.14%	69.32%	75.34%
	LSTM+CNN	59.62%	65.51%	70.32%	74.54%
	Proposed model	77.44%	78.87%	75.63%	76.54%

Table 8 Evaluation comparison between proposed and deep learning model using Fasttext embedding

Word embedding	Models	Precision	F-score	Recall	Accuracy
	CNN	75.23%	78.63%	80.51%	79.75%
	LSTM	81.21%	80.65%	82.54%	82.35%
Fasttext	CNN+LSTM	83.74%	82.52%	79.14%	83.75%
	LSTM+CNN	80.49%	83.14%	82.69%	85.28%
	Proposed model	82.54%	78.12%	83.57%	84.67%

their work and achieved highest 88.72% accuracy. Highest accuracy of 97.67% is obtained by the our proposed hybrid model with enhanced word vector space.

Table 7 presents a comparative analysis of various models employing Word2vec embedding. Precision, F-score, recall, and accuracy metrics are used to evaluate the performance of these models. The model utilizing Word2vec embedding coupled with a CNN architecture achieved a precision of 77.45%, an F-score of 70.25%, a recall of 71.63%, and an accuracy of 71.46%. Interestingly, employing an LSTM model with Word2vec yielded a slightly higher precision at 78.65% but a marginally lower F-score and recall compared to the CNN-Word2vec model. The CNN+LSTM combined model demonstrated the highest precision of 80.23% and an impressive F-score of 79.14%, yet it showed a lower recall and accuracy compared to the previous models. However, when the sequence of models was altered, such as LSTM+CNN, the precision remained at 59.62%, noticeably lower than previous configurations, while the F-score and recall were slightly higher. This suggests a significant impact on precision with the change in the model sequence. The proposed, CNN-LSTM+LSTM-CNN model, displayed a precision of 77.44% and an F-score of 78.87%, with a higher recall compared to other models. Notably, this configuration showed a moderate improvement in accuracy compared to the LSTM+CNN model.

Table 8 shows the Fasttext embedding with CNN architecture achieves a precision of 75.23%, an F-score of 78.63%, a recall rate of 80.51%, and an accuracy of 79.75%. This model's effectiveness is further enhanced when employing the LSTM architecture, demonstrating substantial improvements across all metrics-precision (81.21%), F-score (80.65%), recall (82.54%), and accuracy (82.35%). The subsequent models explore combinations of CNN and LSTM architectures. The CNN+LSTM model showcases an increase in precision (83.74%) and F-score (82.52%), albeit a slight decrease in recall (79.14%) compared to the standalone LSTM model. Conversely, the LSTM+CNN model exhibits slightly lower precision (80.49%) but excels in F-score (83.14%) and recall (82.69%), achieving a notably higher accuracy of 85.28%. The proposed model denoted as CNN-LSTM+LSTM-CNN, yields a precision of 82.54%, an F-score of 78.12%, a recall of 83.57%, and an accuracy of 84.67%.

Table 9 Evaluation comparison between proposed and deep learning model using Glove embedding

Word embedding	Models	Precision	F-score	Recall	Accuracy
	CNN	79.32%	81.48%	78.92%	82.42%
	LSTM	61.00%	81.25% 7	9.67%	80.40%
Glove	CNN+LSTM	76.36%	80.72%	83.56%	82.54%
	LSTM+CNN	82.35%	74.36%	82.89%	83.56%
	Proposed model	84.11%	80.23%	81.84%	83.62%

Table 10 Evaluation comparison between proposed and deep learning model using ELMo embedding

Word embedding	Models	Precision	F-score	Recall	Accuracy
	CNN	77.31%	74.41%	79.11%	80.32%
	LSTM	80.75%	79.62%	81.45%	83.45%
ELMo	CNN+LSTM	80.23%	80.78%	82.56%	84.65%
	LSTM+CNN	79.63%	82.54%	80.78%	83.91%
	Proposed model	81.70%	83.45%	80.19%	84.25%

Table 9 presents a comparative analysis of Glove embedding techniques and their performance metrics in conjunction with various models for sentiment analysis. CNN Performs well with Glove and achieved decent accuracy (82.42%) but lower recall (78.92%) compared to other models. However, its performance varies notably when combined with different models. For instance, with the "LSTM" model, the precision drops to 61%, while the F-score remains high at 81.25%. The recall and accuracy show values of 67% and 80.40%, respectively. Combines CNN's pattern recognition with LSTM's resulting in moderate improvement in F-score (80.72%) compared to standalone CNN but not significantly better accuracy. the LSTM+CNN exhibits improved precision (82.35%) but comparatively lower F-score (74.36%) than the "CNN+LSTM" arrangement. The recall and accuracy for "LSTM+CNN" are 82.89% and 83.56%, respectively. CNN-LSTM+LSTM-CNN configuration presents the highest precision (84.11%) among the listed combinations.

Table 10 demonstrates that using ELMo embeddings with hybrid CNN-LSTM models generally yields the highest accuracy, F-score, and precision, reaching up to 84.25% accuracy, 83.45% F-score, and 81.70% precision for the CNN-LSTM+LSTM-CNN combination, while LSTM models achieve the highest recall at 82.56%. Overall, hybrid models outperform single CNN or LSTM architectures, suggesting that combining their strengths effectively captures both local and long-term dependencies in the data. Table 11 illustrates the performance metrics of different models combined with the "Enhanced word vector space" embedding technique for sentiment analysis. When paired with individual models, the "CNN" model exhibits a precision of 84.24%, an F-score of 83.65%, a recall of 85.78%, and an accuracy of 86.54%, while the "LSTM" model showcases higher precision (84.94%) and F-score (87.78%) but a lower recall (83.44%) and accuracy (88.14%). Combining both models as "CNN+LSTM" or "LSTM+CNN" enhances precision and recall, with "CNN+LSTM" achieving a precision of 87.26%, an F-score of 84.83%, a recall of 85.36%, and an accuracy of 87.54%. Conversely, "LSTM+CNN" yields a slightly lower precision (84.21%) but a notably higher F-score (88.82%) and similar recall (86.23%) and accuracy (86.32%). Remarkably, the most complex configuration, "CNN-LSTM+LSTM-

Table 11 Evaluation comparison between proposed and deep learning model using enhanced word vector space embedding

Word embedding	Models	Precision	F-score	Recall	Accuracy
	CNN	84.24%	83.65%	85.78%	86.54%
	LSTM	84.94%	87.78%	83.44%	88.14%
Enhanced word vector space	CNN+LSTM	87.26%	84.83%	85.36%	87.54%
	LSTM+CNN	84.21%	88.82%	86.23%	86.32%
	Proposed model	89.65%	90.16%	91.11%	91.64%

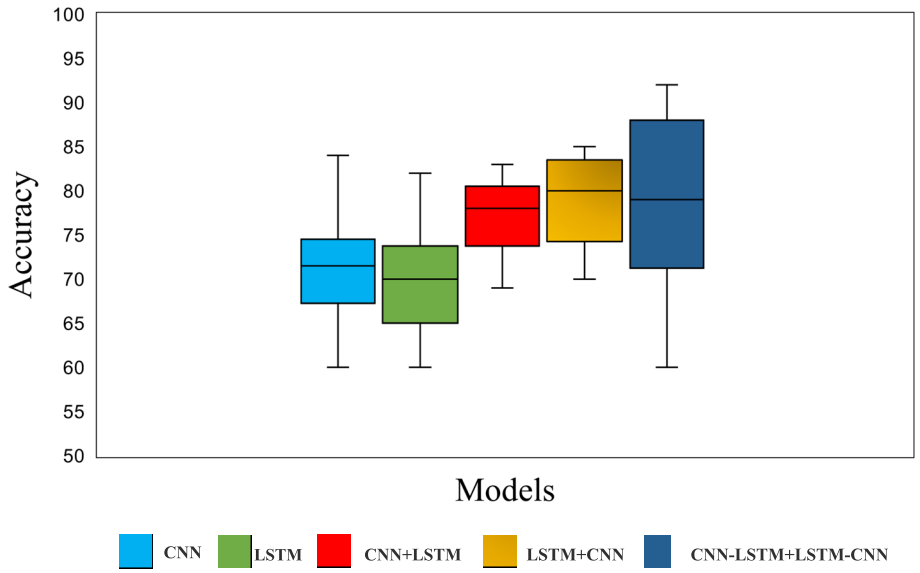


Fig. 7 Deep learning versus proposed model accuracy comparison

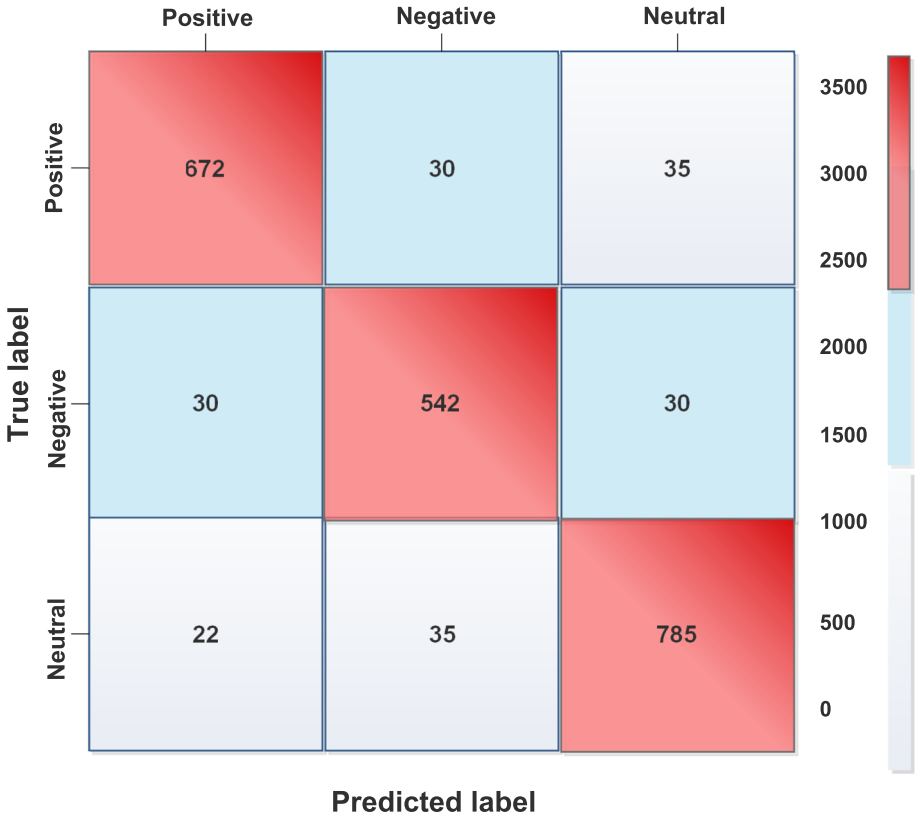


Fig. 8 Confusion matrix of the proposed classification model

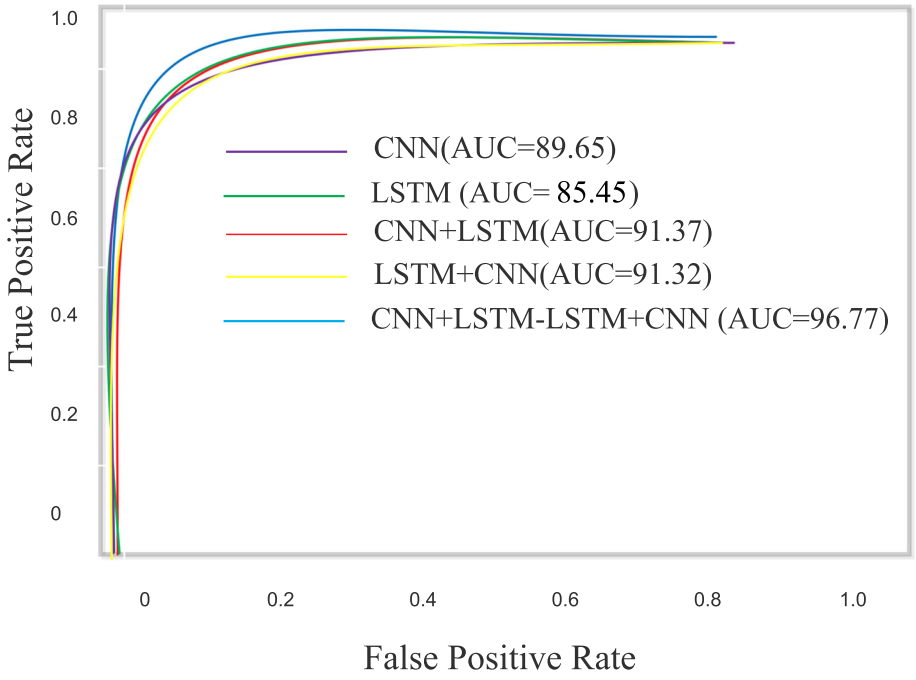


Fig. 9 ROC curve for all classification model

CNN," demonstrates the highest precision (89.65%), F-score (90.16%), recall (91.11%), and accuracy (91.64%) among the listed combinations, highlighting its superior performance in sentiment classification within the context of analyzed Hindi text related to the COVID-19 pandemic. Tables 7, 8, 9, 10, 11 demonstrates that the Enhanced Word Vector Space consistently outperforms other embedding techniques across all model architectures, achieving the highest accuracy, F-score, precision, and recall in most cases. Hybrid CNN-LSTM models generally perform better than single CNN or LSTM models, with the CNN-LSTM+LSTM-CNN configuration achieving the best results for all embeddings.

The accuracy acquired by the suggested model is shown as a boxplot in Fig. 7. Based on the analysis, it is found that the proposed model with enhanced word vector space provides the highest accuracy. Boxplot signifies that the performance of the suggested model is consistent.

The proposed model's confusion matrix is shown in Fig. 8. It demonstrated that the neutral feeling predominates over both positive and negative feelings.

AUC evaluates how well a model predicts the feature. Region with a range of 1 indicates a successful test, while region with a 0 indicates that the prediction model incorrectly categorized every event (Fig. 9).

5 Conclusion

This work suggested an upgraded word embedding-based deep learning model for emotion classification of Hindi tweets regarding COVID-19. Enhanced word vector space embedding technique is developed by utilizing natural language processing methods, lexicon-based methodologies, POS tagging, Word2vec, Elmo, and Fasttext to extract feature vector from

collected tweets and to enhance precision of embedded vectors. Ensemble Hybrid model of CNN-LSTM + LSTM-CNN is used to classify the sentiments and highest accuracy of **91.64%**. The experimental findings proved that the proposed model is more successful for evaluating Hindi tweets sentiment in comparison with existing models.

Impact and potential applications of our proposed solution Our hybrid deep learning model for COVID-19 Hindi sentiment analysis can have a significant impact on several areas:

- Public health surveillance: By analyzing public sentiment on social media, our model provides valuable insights into public concerns, anxieties, and perceptions of the pandemic. Using this information, public health messaging, resource allocation, and intervention strategies can be improved.
- Mental health support: Understanding public sentiment can identify those who are stressed or anxious over the pandemic. Using this information, mental health support services and interventions can be targeted more effectively.
- Social media analysis: Our model can be used by social media platforms and researchers to analyze large volumes of user-generated COVID-19 content. This can help detect misinformation, understand public opinion trends, and track the pandemic's evolution.
- Government agencies: Public health departments and disaster relief organizations can use the model to track public sentiment and tailor their response strategies.
- Non-profit organizations: Mental health organizations can use the model to identify communities in need of support and target their outreach efforts.

Limitations

Following are the constraints and limitations associated with our methodology.

- Data Limitations: One of the primary limitations pertains to the availability and nature of the dataset used. Despite the extensive collection of Hindi tweets, limitations related to data biases, representativeness, and coverage might exist, influencing the model's generalizability to diverse contexts.
- Language and Contextual Challenges: Another aspect we've highlighted involves the inherent complexities of sentiment analysis for Hindi text, including language nuances, dialectical variations, and contextual ambiguities specific to social media language, which might pose challenges in accurate sentiment categorization.
- Model Complexity and Resource Intensiveness: The complexity of the hybrid model, which integrates multiple word embeddings and employs deep learning techniques, may present challenges in computational resource requirements, making it less feasible for resource-constrained environments or real-time applications.

In future, sarcasm and irony text will be included for analysis. The post-level and domain-specific ontology contextual information of words and ablation study will also be investigated.

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Data Availability This study used the Python library to collect tweets from Twitter.

Declarations

Competing interest statement There is no conflict of interest.

Ethical approval No study with human participants has been conducted.

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